INTRODUCTION TO COMPUTING

MODULE 1

INTRODUCTION TO THE PROFESSION

Objectives

- Understand the different knowledge areas in computing
- Describe career opportunities available in various segment of the computer industry.
- Understand the different jobs available in an IT department.
- Identify the benefits of computing-related certification for employers,

Objectives (cont)

- Identify the different areas in IT certification.
- Understand and be aware of different Emerging Technologies

INTRODUCTION TO THE COMPUTING INDUSTRY

What is Computer Science?

- Ranges from theory through programming to cutting-edge development of computing solutions.
- Offers a foundation that permits graduates to adapt to new technologies and new ideas.
- Most computer science programs require some mathematical background.

What is Computer Science? (cont)

The work of computer scientists falls into three categories:

a. designing and building software
b. developing effective ways to solve computing problems, such as storing information in databases, sending data over networks or providing new approaches to security problems
c. devising new and better ways of using computers and addressing particular challenges in areas such as robotics, computer vision, or digital forensics

What is Information Systems?

- concerned with the information that computer systems can provide to aid a company, non-profit or governmental organization in defining and achieving its goals.
- also concerned with the processes that an enterprise can implement and improve using information technology

What is Information Systems? (cont)

IS professionals must:

a. understand both technical and organizational factors
b. be able to help an organization determine how information and technology enabled business processes can provide a foundation for superior organizational performance

What is Information Technology?

- Information technology (IT) is a label that has two meanings. In common usage, the term “information technology” is often used to refer to all of computing.
- As a name of an undergraduate degree program, it refers to the preparation of students to meet the computer technology needs of business, government, healthcare, schools, and other kinds of organizations.

What is Information Technology? (cont)

- IT professionals possess the right combination of knowledge and practical, hands-on expertise to take care of both an organization's information technology infrastructure and the people who use it.
- They assume responsibility for selecting hardware and software products appropriate for an organization.

What is Information Technology? (cont)

- They integrate those products with organizational needs and infrastructure, and install, customize and maintain those applications, thereby providing a secure and effective environment that supports the activities of the organization's computer users.

What is Information Technology? (cont)

IT discipline require special skills such as:

- Understanding how networked systems are composed and structured, and what their strengths and weaknesses are.
- Important software systems concerns such as reliability, security, usability, and effectiveness and efficiency for their intended purpose

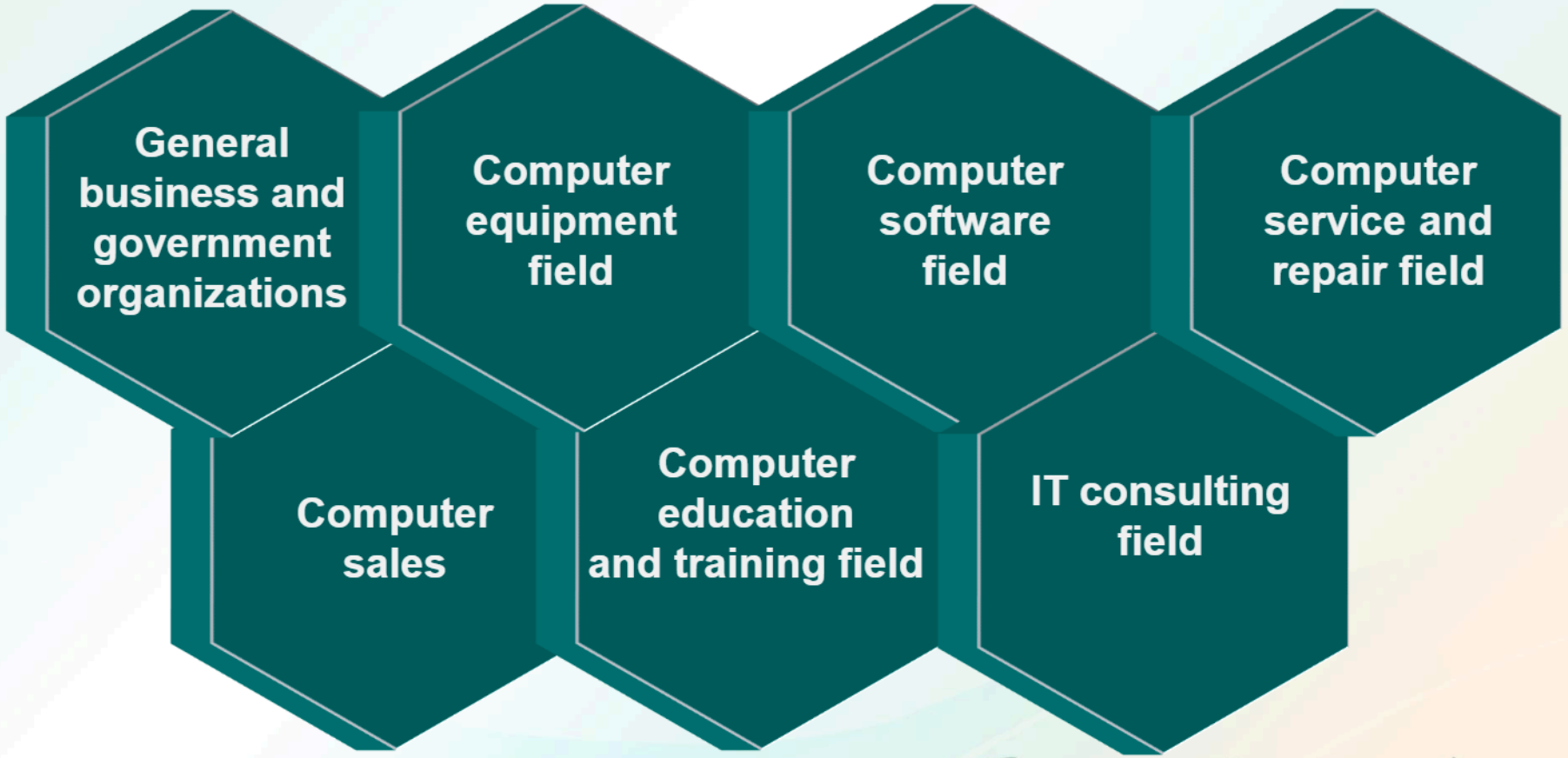
The Computer Industry

What is the expected demand for computer professionals?

- Currently 10.5 million computer workers increase of 2 to 3% annually
- The number of students majoring in computer-related fields is declining

Careers in the Computer Industry

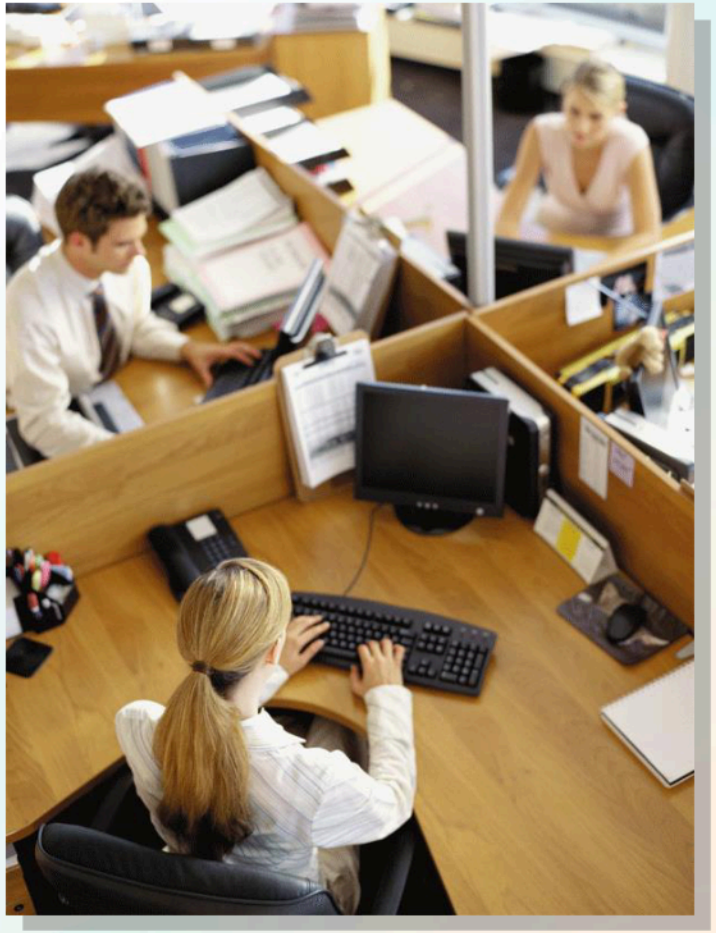
What are the primary areas where job opportunities are found?



Careers in the Computer Industry (cont)

What is an information technology (IT) department?

- Department in business or government agency that employs people in computer-related jobs
- Responsible for keeping all computer operations and networks operating smoothly



Careers in the Computer Industry (cont)

What jobs are available in an IT department?

Management	Chief Information Officer / VP of IT	E-commerce administrator	Project leader / manager
	Chief Security Officer	Network administrator	
System development and programming	Application programmer	Software engineer	Technical writer
	Computer scientist	Systems analyst	Technical lead
	Database analyst	Systems programmer	Web page author
Technical services	Computer forensics specialist	Desktop publisher	Quality assurance specialist
	Computer technician	Graphic designer / illustrator	Security administrator
	Database administrator	Network security specialist	Web administrator / Webmaster
Operations	Computer operator	Data communications analyst	
Training	Corporate trainer	Help desk specialist	

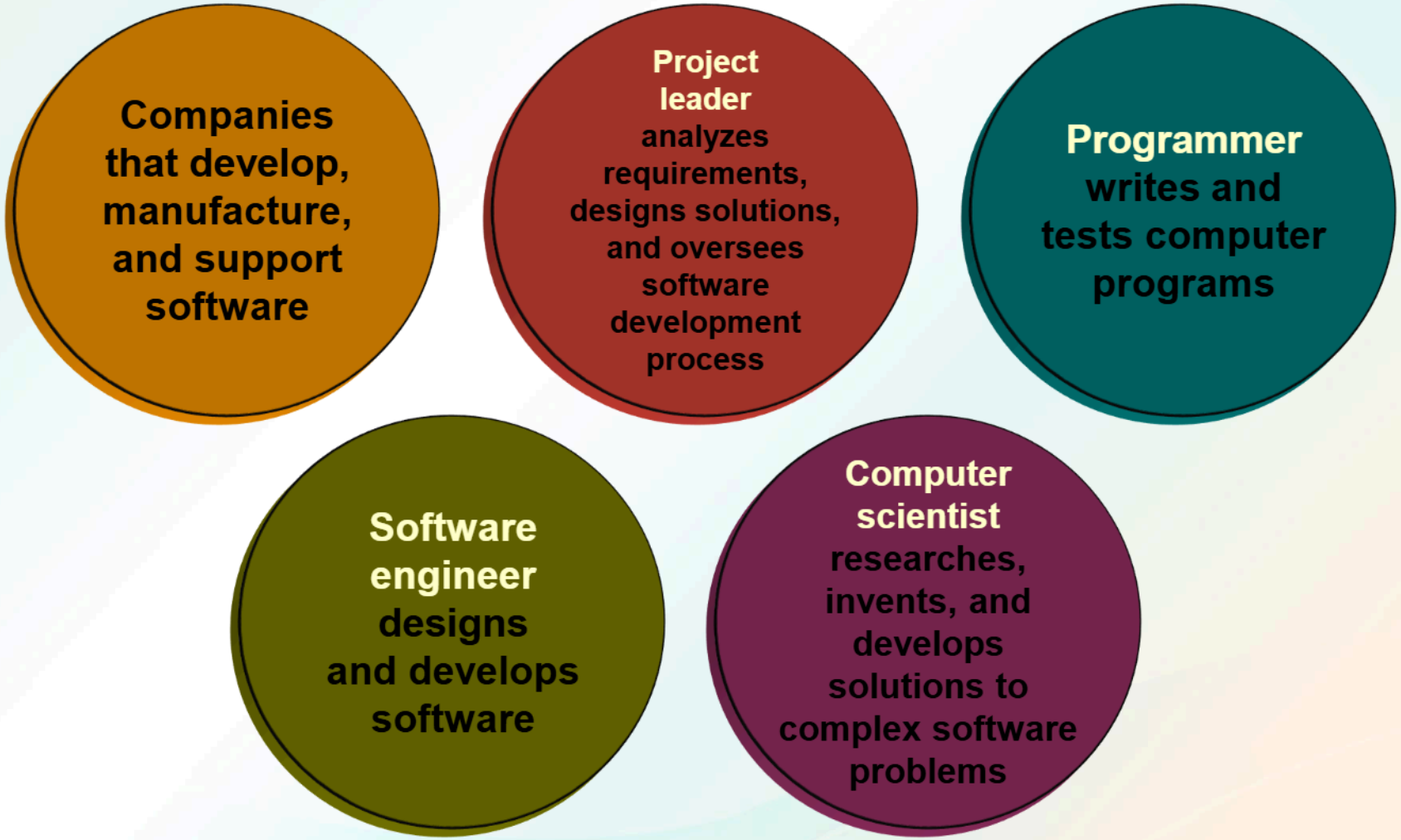
Careers in the Computer Industry (cont)

What is the computer equipment field?

- Manufacturers and distributors of computers and computer-related hardware.
- Careers involve design, manufacture, and production of computer equipment

Careers in the Computer Industry (cont)

What is the computer software field?



Careers in the Computer Industry (cont)

What is the computer service and repair field?

- Provides preventative maintenance, component installation, and repair services
- Computer technician—installs, maintains, and repairs hardware

Careers in the Computer Industry (cont)

What are career opportunities in computer sales?

- Computer sales people need general understanding of computers and specific knowledge of the product they are selling

Careers in the Computer Industry (cont)

What are career opportunities in education and training?

- Schools, colleges, universities, and private companies need educators and trainers.
- Corporate trainers teach employees how to use software, design and develop systems, and program

Careers in the Computer Industry (cont)

What is an IT consultant?

Individual who draws upon expertise in specialized area of computers

- Computer selection
- System design
- Communications
- Hardware and software configuration
- Network design and installation
- Web development

Preparing for a Career in the Computer Industry

What are the three broad disciplines in higher education that produce the majority of entry-level employees in the computer industry?

COMPUTER DISCIPLINE DIFFERENCES

Computer Information Systems*	Computer Science/ Software Engineering	Computer Engineering
Practical and application oriented	Theory oriented	Design oriented
Business and management oriented	Mathematics and science oriented	Mathematics and science oriented
Understand how to design and implement information systems	Understand the fundamental nature of hardware and software	Understand the fundamental nature of hardware and electronics
Certificates Degrees include A.A., A.A.S., A.S., B.A., B.S., M.S., Ph.D	Degrees include B.S., M.S., Ph.D.	Degrees include B.S., M.S., Ph.D.

Preparing for a Career in the Computer Industry (cont)

What is computer information systems (CIS)?

- Programs that emphasize technical knowledge and skills
- Sometimes called management information systems (MIS) or management information technology

Computer Courses:

- Introduction to Computers
- Object-Oriented Programming
- Networks
- Computer Operating Systems
- Systems Analysis and Design
- Database Management Systems
- Web Development
- Two semesters of two or more of the following programming languages: C++, Java, Visual Basic .NET 2003, COBOL

Other Courses:

- Algebra
- Accounting and other business courses
- Statistics

Preparing for a Career in the Computer Industry (cont)

What is computer science (CS)?

- Programs that stress theoretical side of programming and operating systems
- Also called software engineering.

Computer Courses:

- Introduction to Computers
- Object-Oriented Programming
- Data Communications
- Systems Analysis and Design
- Software Engineering
- Data and File Structures
- Computer Operating Systems
- Artificial Intelligence
- Two semesters of at least one of the following programming languages: C++, Java, Assembler

Math Courses:

- Calculus – 2 semesters
- Physics – 2 semesters

Preparing for a Career in the Computer Industry (cont)

What is computer engineering (CE)?

- Teaches students how to design and develop the electrical components found in computers and peripheral devices.
- Required courses include application and systems programming, data and file structures, networks, and operating systems.

Preparing for a Career in the Computer Industry (cont)

How do companies advertise computer-related job openings?

- Some Web sites specialized in helping people find jobs.
- Some allow job seekers to submit resumes online

Preparing for a Career in the Computer Industry (cont)

PROFESSIONAL COMPUTER ORGANIZATIONS
Association for Computing Machinery (ACM)
Association for Information Systems (AIS)
Association for Women in Computing (AWC)
Association of Information Technology Professionals (AITP) (formerly DPMA)
IEEE Computer Society
Independent Computer Consultants Association (ICCA)
Information Systems Security Association (ISSA)
Institute of Electrical and Electronics Engineers (IEEE)
National Association of Programmers (NAP)

What are some organizations for computer professionals?

- Association for Computing Machinery (ACM)—science-oriented
- Association of Information Technology Professionals (AITP)—programmers, systems analysts, and information processing managers

Preparing for a Career in the Computer Industry (cont)

What is a user group?

- **Collection of people with common computer equipment or software interests**

EXAMPLES OF USER GROUPS
Apple User Groups
Association of Personal Computer User Groups (APCUG)
C/C++ Users Group (CUG)
Independent Oracle Users Group (IOUG)
Java Users Groups (JUG)
Linux Users Groups WorldWide (LUGWW)
.NET User Groups
UNIX User Groups

Preparing for a Career in the Computer Industry (cont)

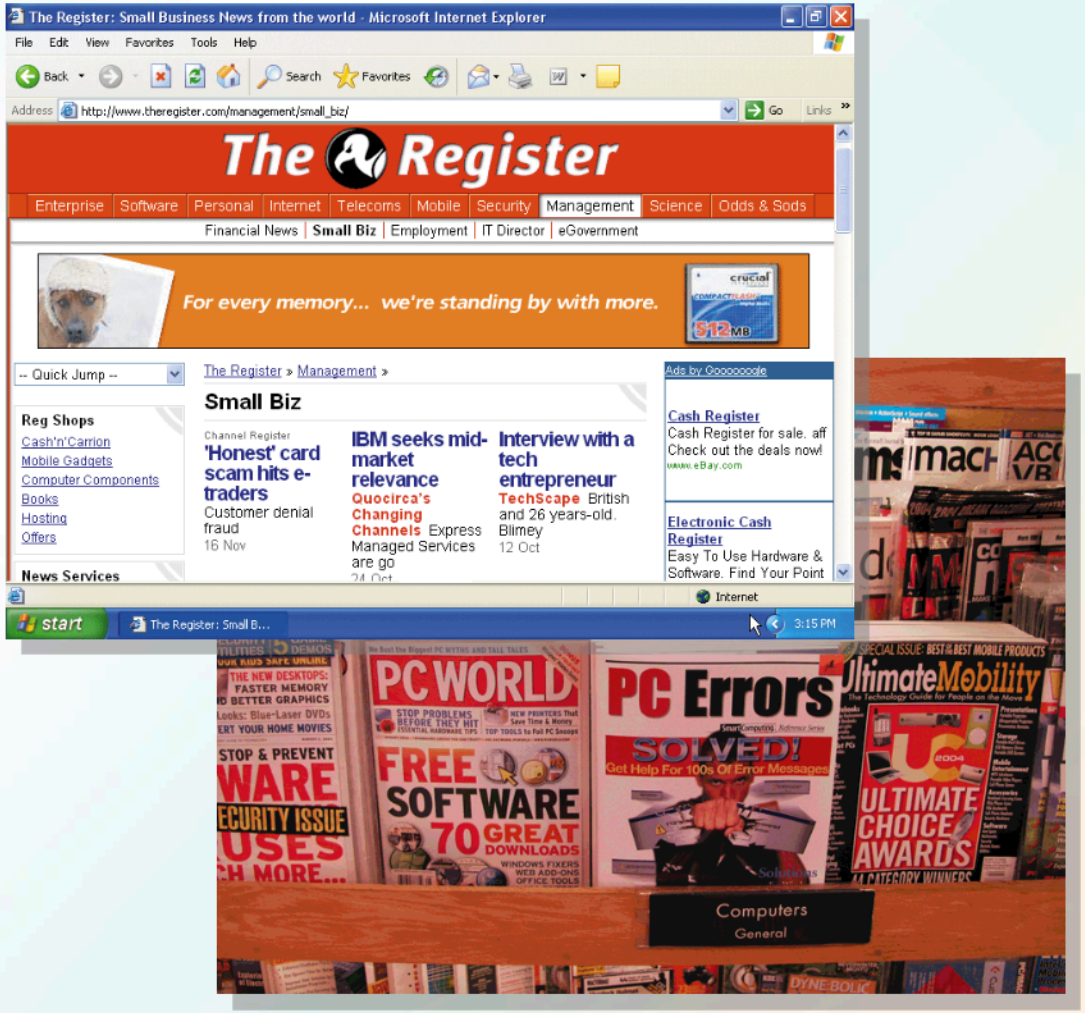
What is professional growth and continuing education?

- Events such as workshops, seminars, conferences, conventions, and trade shows
  - CES—trade show that brings together hundreds of vendors and more than 130,000 attendees
- Training in form of books, video-based training, computer-based training (CBT), Web-based training (WBT), and instructor-led training

Preparing for a Career in the Computer Industry (cont)

How can you stay informed about the computer industry?

- Computer industry publications
- News, blogs, and technical Web sites



CERTIFICATION AND TECHNOLOGY TRENDS

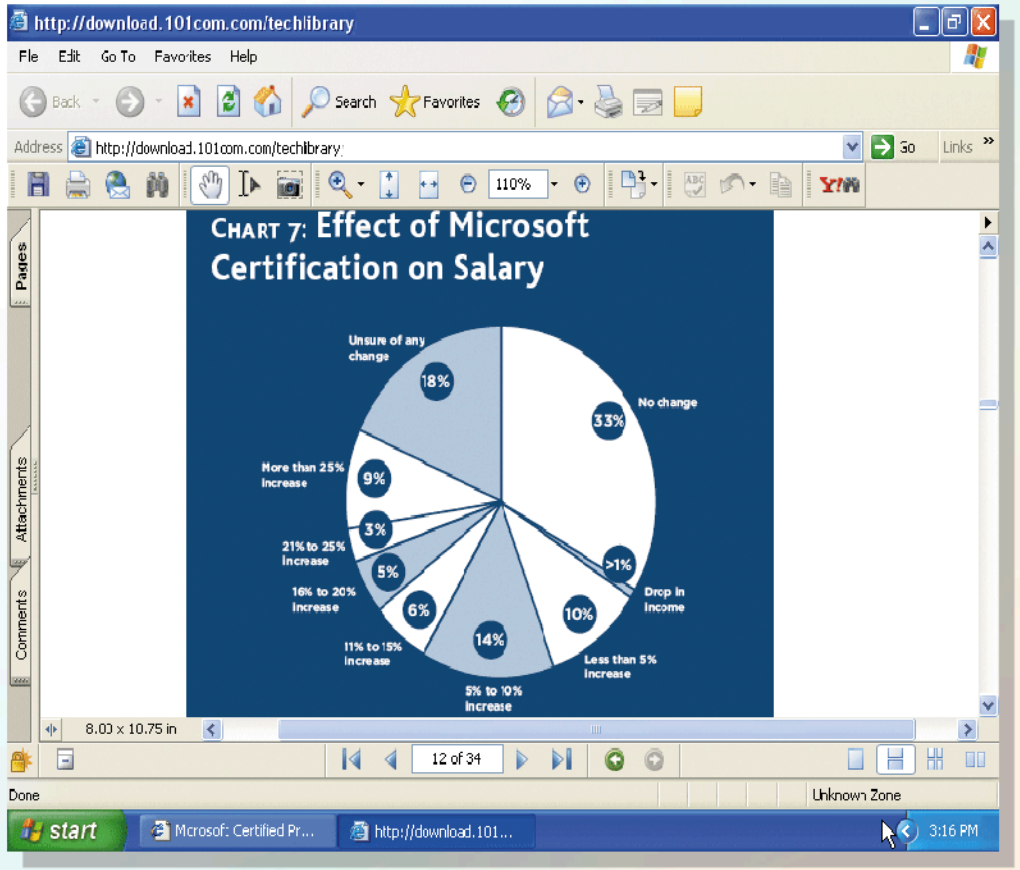
Certification

What is certification?

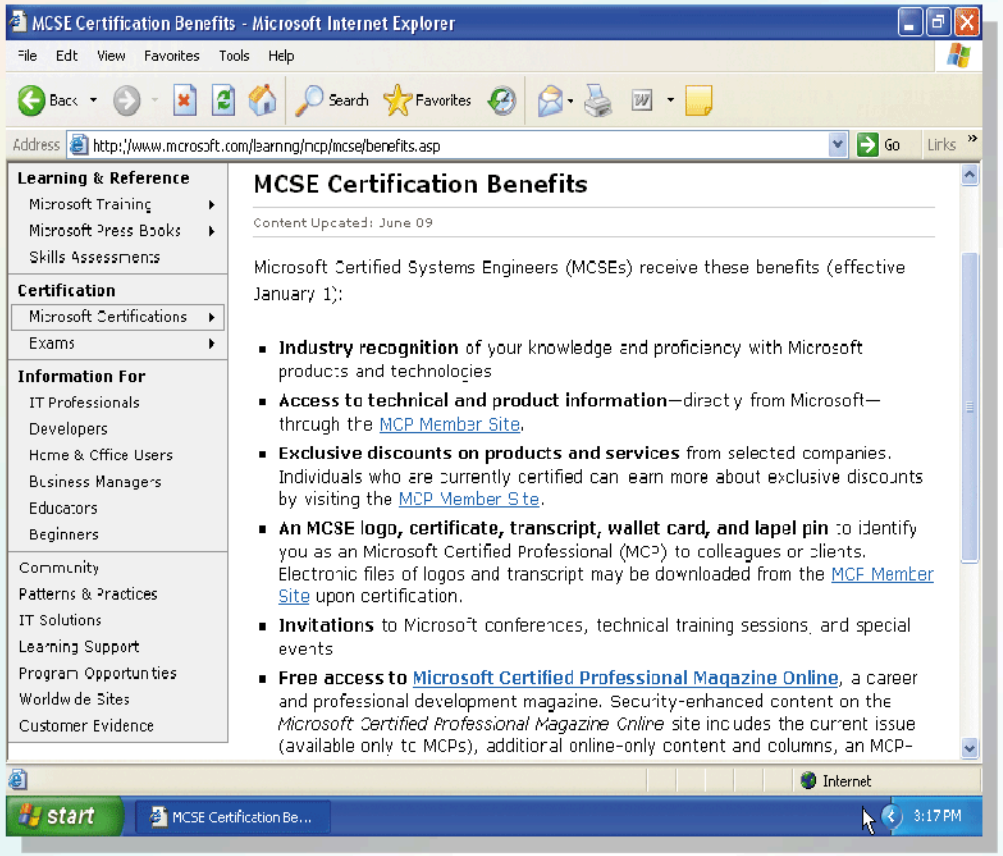
- Process of verifying technical knowledge in a particular area.
- Proficiency is acknowledged with certificate.

Certification (cont)

- Enhances employee's career, provides better standing as industry professional, and higher salaries and bonuses



Certification (cont)



- Vendors provide access to technical information, advanced training, and discounts on publications and tools

Certification (cont)

How does certification benefit customers, employers, and industry?

Certification (cont)

What are factors to consider in choosing a certification?

FACTORS TO CONSIDER IN SELECTING A CERTIFICATION

- Consider the expenses and the time involved to obtain the certification
- Examine employment projections
- Look at job listings to see what certifications are sought
- Read evaluations of certifications
- Talk to people in the industry
- Think about complementary combinations of certifications to meet your goals

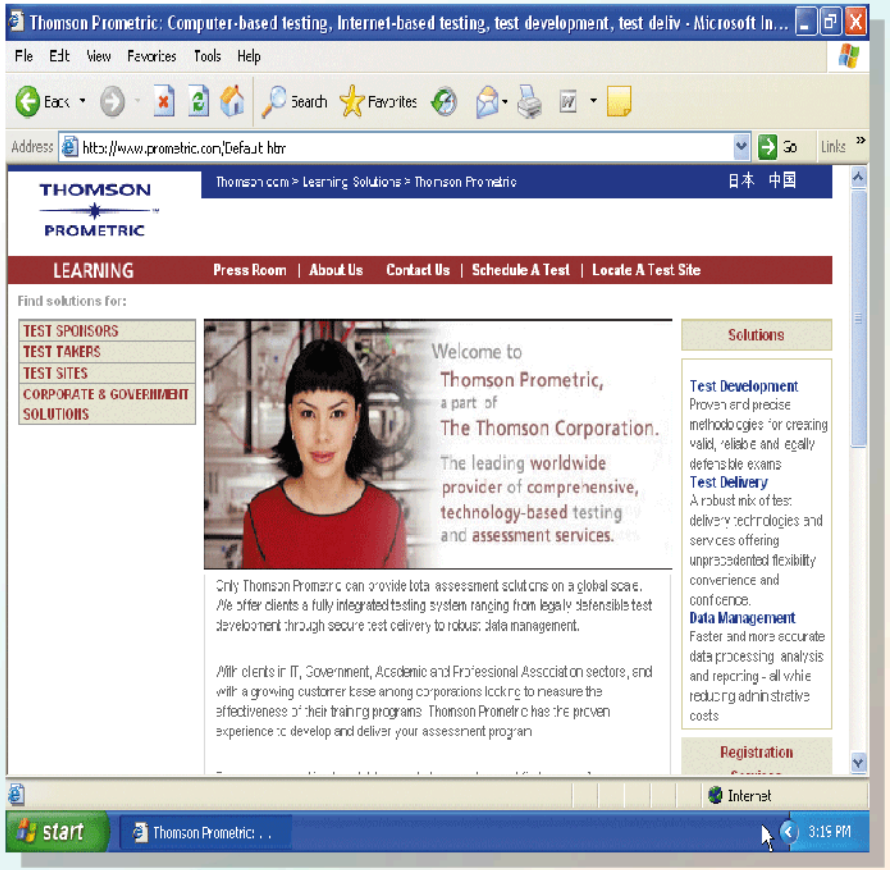
Certification (cont)

What options are available to prepare for certification?

Certification (cont)

What are examinations like?

- Taken using computers
- Some are multiple choice, others are skill based
- With computerized adaptive testing, the tests analyze a person's responses while taking the test



A Guide to Certification

What are application software certifications?

Focus on knowledge of application software

Sometimes called end-user certifications

- Microsoft Office Specialist (MOS) sponsored by Microsoft
- Microsoft Certified Desktop Support Technician (MCDST) sponsored by Microsoft
- Adobe Certified Expert sponsored by Adobe Systems
- IBM Certified Professional for Lotus Software sponsored by IBM
- Macromedia Certified Professional sponsored by Macromedia

A Guide to Certification (cont)

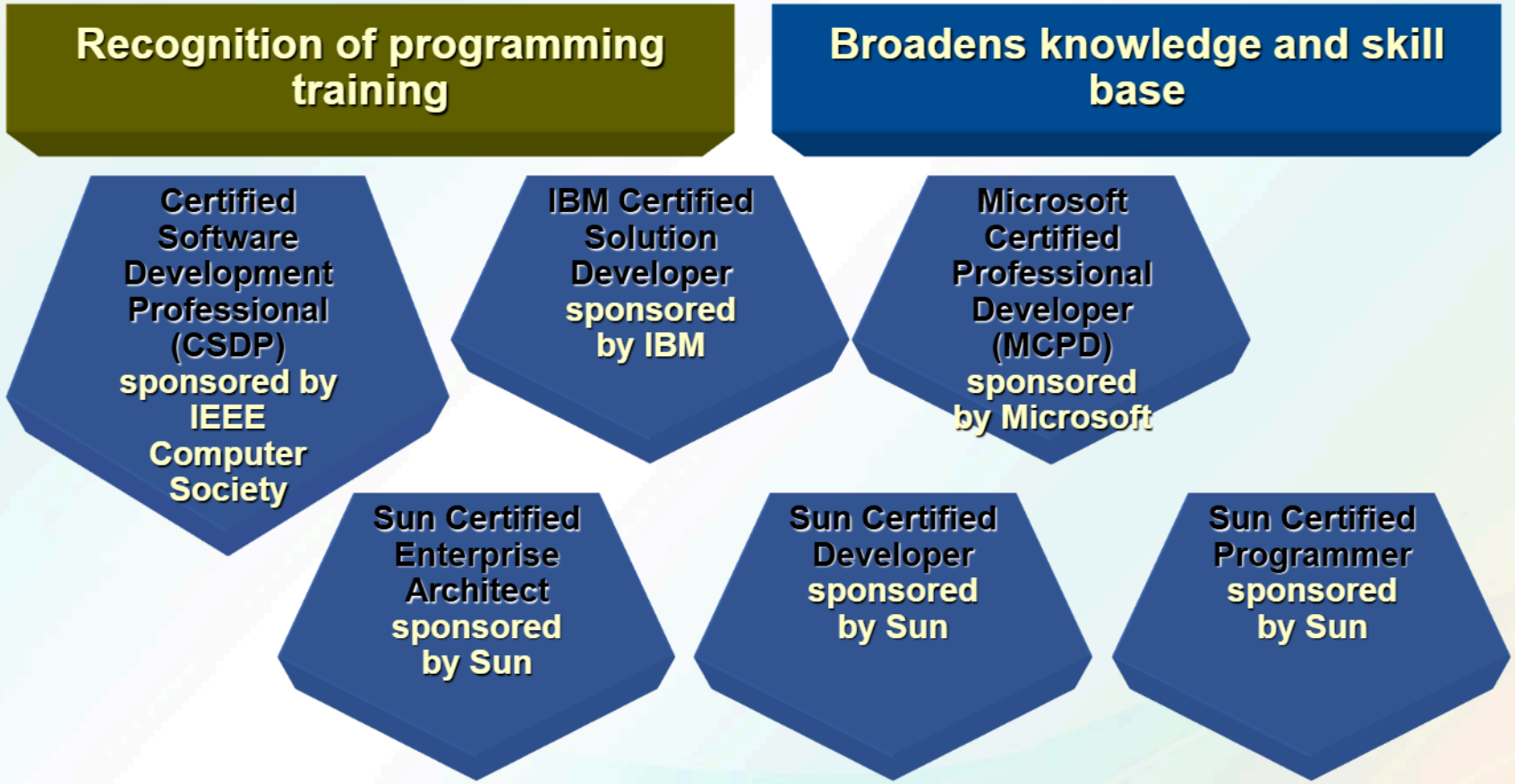
What are operating system certifications?

Focus on particular skills—user, operator, system administrator, and system engineer

Differing levels of certification

A Guide to Certification (cont)

What are programming certifications?



A Guide to Certification (cont)

What are hardware certifications?

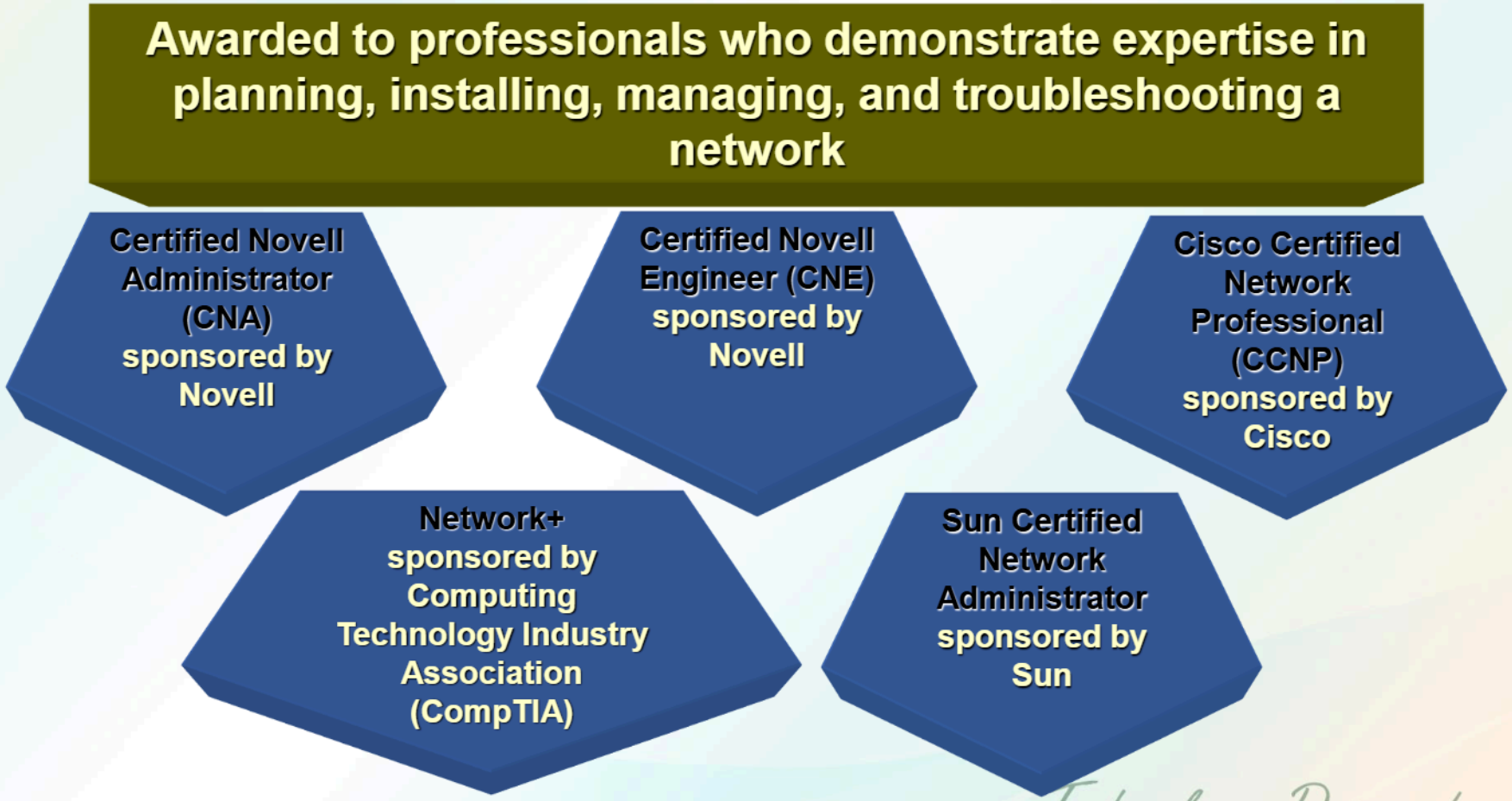
Recognition of general knowledge of hardware and controlling software needed to set up, maintain, and repair a computer

Can be narrowly or broadly focused

- A+ sponsored by Computing Technology Industry Association (CompTIA)
- IBM eServer Certified Specialist sponsored by IBM
- NACSE Network Technician (NNT) sponsored by National Association of Communication Systems Engineers (NACSE)

A Guide to Certification (cont)

What are networking certifications?



A Guide to Certification (cont)

What are computer forensics certifications?

Discovery, collection, and analysis of evidence found on computers and networks

Certificate holders are considered highly credible during computer crimes cases

- Certified Electronic Evidence Collection Specialist (CEECS) sponsored by International Association of Computer Investigative Specialists (IACIS)
- Certified Forensic Computer Examiner (CFCE) sponsored by International Association of Computer Investigative Specialists (IACIS)
- Certified Computer Examiner (CCE) sponsored by International Society of Forensic Computer Examiners (ISFCE)

A Guide to Certification (cont)

What are security certifications?

Measure a candidate's ability to identify and control security risks to computer resources

- Certified Information Security Professional (CISSP) sponsored by $(ISC)^2$
- Systems Security Certified Practitioner (SSCP) sponsored by $(ISC)^2$
- Security Certified Network Professional (SCNP) sponsored by The Security Certified Program
- Security Certified Network Architect (SCNA) sponsored by The Security Certified Program

A Guide to Certification (cont)

What are Internet certifications?

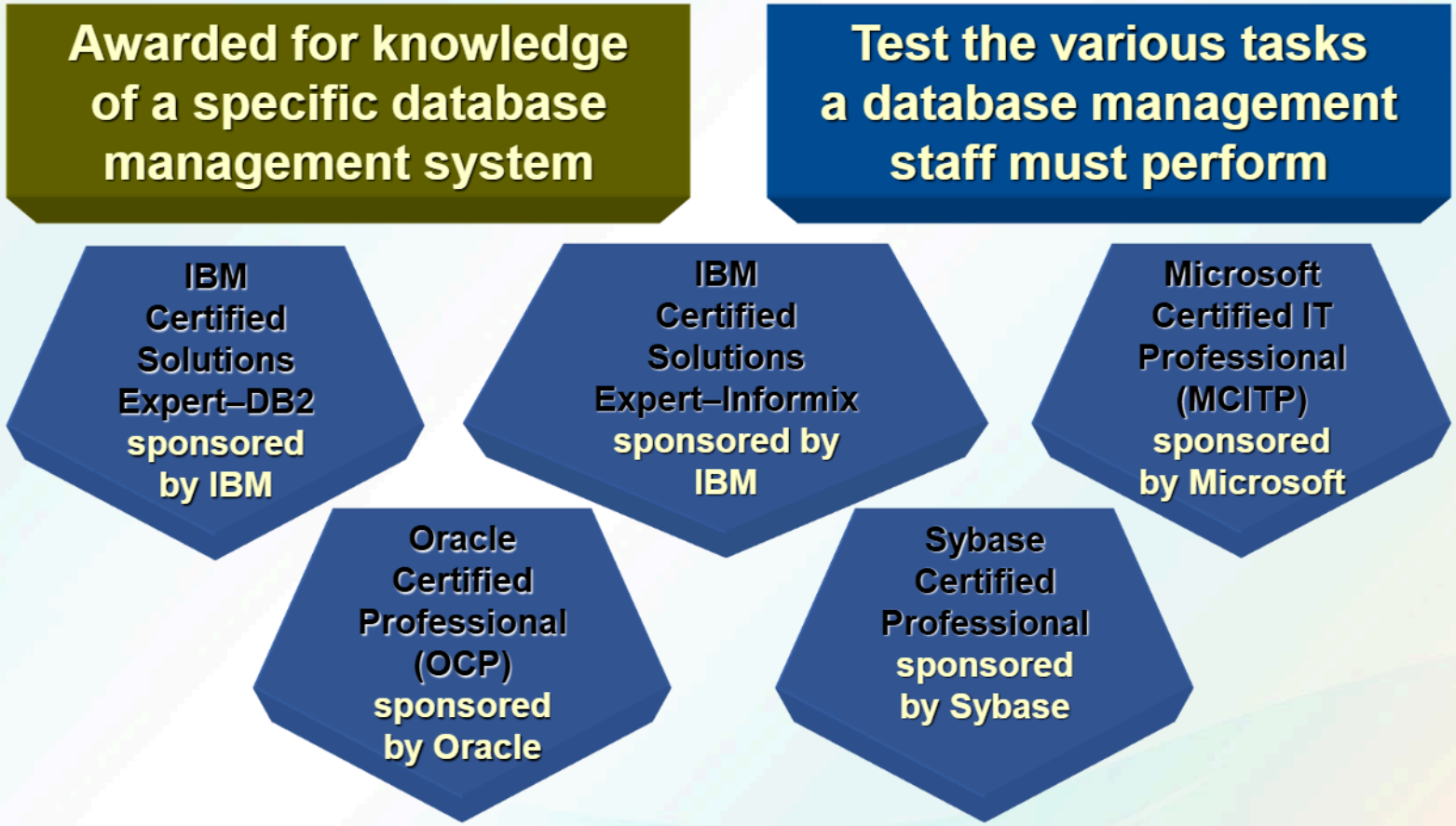
Require technical expertise in networking hardware and configuration

Include Web management, Web programming, and Web development certifications

- Certified Internet Webmaster (CIW) sponsored by Prosoft Training
- Cisco Certified Internetwork Expert (CCIE) sponsored by Cisco
- Certified Web Professional (CWP) sponsored by International Webmasters Association

A Guide to Certification (cont)

What are database system certifications?



Latest Trends and Technologies

Artificial Intelligence (AI)

- Artificial Intelligence continues to be a trend to watch because its effects on how we live, work and play are only in the early stages.
- AI refers to computers systems built to mimic human intelligence and perform tasks such as recognition of images, speech or patterns and decision making.
- AI can do these tasks faster and more accurately than humans.

Latest Trends and Technologies (cont)

Machine Learning

- Machine Learning is a subset of AI.
- Computers are programmed to learn to do something they are not programmed to do.
- They learn by discovering patterns and insights from data.
- Neural networks, natural language processing (NLP), and deep learning are some subsets of Machine Learning.

Latest Trends and Technologies (cont)

Robotic Process Automation or RPA

- RPA is the use of software to automate business processes such as interpreting applications, processing transactions, dealing with data, and even replying to emails.
- RPA automates repetitive tasks that people used to do.
- These are not just the menial tasks of a low-paid worker: up to 45 percent of the activities we do can be automated, including the work of financial managers, doctors and CEOs.

Latest Trends and Technologies (cont)

Blockchain

- Most people think of blockchain technology in relation to cryptocurrencies such as Bitcoin.
- Blockchain offers security that is useful in many other ways. In the simplest of terms, blockchain can be described as data you can only add to, not take away from or change.
- Blockchains are consensus-driven so no one entity can take control of the data.

Latest Trends and Technologies (cont)

Edge Computing

- Cloud computing has become mainstream, with major players AWS (Amazon Web Services), Microsoft Azure and Google Cloud dominating the market.
- Cloud computing is no longer the emerging technology. Edge is.
- Edge computing is designed to help solve some problems of cloud computing as a way to bypass the latency caused by cloud computing and getting data to a data center for processing.

Latest Trends and Technologies (cont)

Virtual Reality (VR) and Augmented Reality (AR)

- Virtual Reality (VR) immerses the user in an environment.
- Augment Reality (AR) enhances their environment.
- Although VR has primarily been used for gaming thus far, it has also been used for training.
- Both have enormous potential in training, entertainment, education, marketing, and even rehabilitation after an injury.

Latest Trends and Technologies (cont)

Cyber Security

- Cyber security might not seem like emerging technology but it is evolving just as other technologies are.
- Hackers will continue to find ways to get through even the toughest security measures.
- New technology is being adapted to enhance security.
- As long as we have hackers, we will have cyber security as an emerging technology because it will constantly evolve to defend against those hackers.

Latest Trends and Technologies (cont)

Internet of Things

- Many “things” are now being built with WiFi connectivity, meaning they can be connected to the Internet—and to each other.
- IoT enables devices, home appliances, cars and much more to be connected to and exchange data over the Internet.
- The number of IoT devices reached 8.4 billion in 2017 and is expected to reach 30 billion devices after 3 years.

REFERENCES

- **Docter, Q., Dulaney, E., & Skandier, T. (2016).** *CompTIA + Complete Study Guide (3rd Edition)*
- **Meyers, M. (2016).** *All In One CompTIA A+ Certification.* **Mc Graw Hill.**
- **Pearson. (2016).** *Introduction to Computers and Information Technology*. **Upper Saddle River, NJ: Pearson.**
- **Tavani, H. T. (2016).** *Ethics and Technology.* **Wiley.**
- **White, R. (2015).** *How Computer Work: The Evolution of Technology*. **IN: Que.**